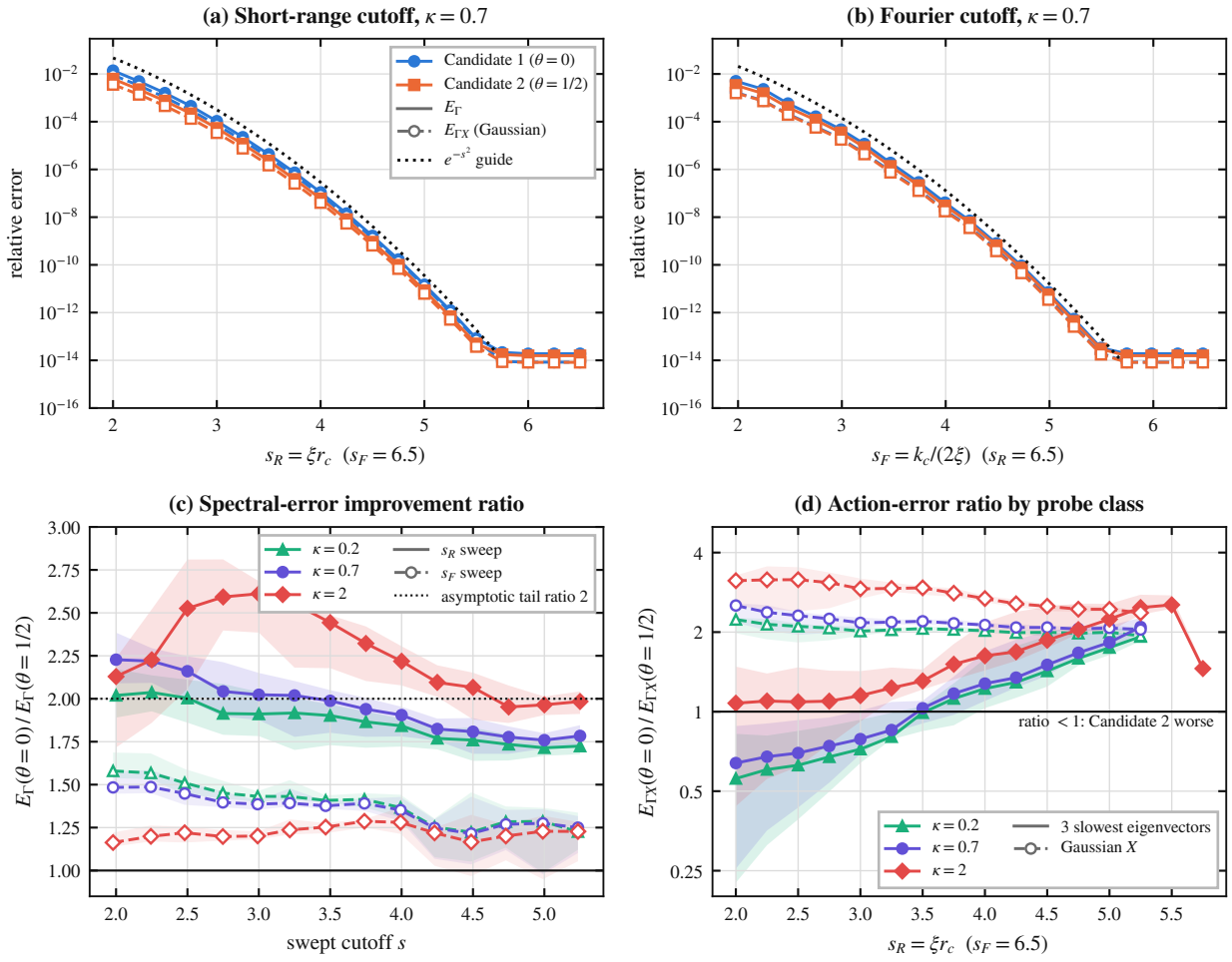
(a) Short-range cutoff, $\kappa = 0.7$
relative error
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
2
3
4
5
6
$s_R = \xi r_c$ $(s_F = 6.5)$
Candidate 1 ($\theta = 0$)
Candidate 2 ($\theta = 1/2$)
E_Γ
$E_{\Gamma X}$ (Gaussian)
e^{-s^2} guide
(b) Fourier cutoff, $\kappa = 0.7$
relative error
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
2
3
4
5
6
$s_F = k_c/(2\xi)$ $(s_R = 6.5)$
(c) Spectral-error improvement ratio
$E_\Gamma(\theta = 0) / E_\Gamma(\theta = 1/2)$
3.00
2.75
2.50
2.25
2.00
1.75
1.50
1.25
1.00
2.0
2.5
3.0
3.5
4.0
4.5
5.0
swept cutoff s
$\kappa = 0.2$
$\kappa = 0.7$
$\kappa = 2$
s_R sweep
s_F sweep
asymptotic tail ratio 2
(d) Action-error ratio by probe class
$E_{\Gamma X}(\theta = 0) / E_{\Gamma X}(\theta = 1/2)$
4
2
1
0.5
0.25
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
$s_R = \xi r_c$ $(s_F = 6.5)$
ratio < 1: Candidate 2 worse
$\kappa = 0.2$
$\kappa = 0.7$
$\kappa = 2$
3 slowest eigenvectors
Gaussian X